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Aim: Implementation of nesC Programming Model

Create a basic nesC application using modules and configurations to understand component wiring and interface binding.

Step 1: The Module (BlinkC.nc)

This component contains only the code logic. It does not know where the timer or LEDs come from; it just declares that it needs them.

CODE:

```
#include "Timer.h"

module BlinkC {
    // Interfaces used by this module
    uses interface Boot;
    uses interface Leds;
    uses interface Timer<TMilli> as Timer0;
}

implementation {
    event void Boot.booted() {
        // Command: start timer
        call Timer0.startPeriodic(500);
    }

    event void Timer0.fired() {
        // Command: toggle LED
        call Leds.led0Toggle();
    }
}
```

Step 2: The Configuration / Wiring (BlinkAppC.nc)

This component has no logic. Its only job is component wiring (binding interfaces together).

Code:

```
configuration BlinkAppC {
}

implementation {
    // 1. Declare the components
    components MainC;
    components LedsC;
    components new TimerMilliC() as Timer0;
    components BlinkC as App;

    // 2. Wire the interfaces (Interface Binding)
    App.Boot -> MainC.Boot;
    App.Leds -> LedsC;
    App.Timer0 -> Timer0;
}
```

```

lrc,TinyScheduler,TaskBasic,TaskBasic,TaskBasic,runT
doNotenumber=sin_node\()\ -fnesc-gcc=gcc -I../tos/
nlcaz/sln -I../tos/platforms/nlcaz -I../tos/platf
-I../tos/chips/cc2420/sln -I../tos/chips/cc2420
csma/sln -I../tos/chips/cc2420/csma -I../tos/chip
-I../tos/chips/cc2420/lopan/sln -I../tos/chips/cc2
c2420/receive/sln -I../tos/chips/cc2420/receive -I
-I../tos/platforms/nlcaz/chips/at45db -I../tos/
atm128/adc -I../tos/chips/atm128/plns/sln -I../tos/
/chips/atm128/timer -I../tos/lib/power -I../tos/
tos/types -I../tos/interfaces -DIDENT_APPNAME="\
IDENT_UIDHASH=0x0e1c6db1 -Wno-nesc-data-race BlinkAp
c-dump-tags -fnesc-dumpfile=app.xml

ction '_nesc_atomic_start' which is not static

ENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -D
tos/lib/tossln -DHAVE_CONFIG_H

rity]

_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -DID
tossln
ENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -D
/lib/tossln

stdc++ -o _TOSSIModule.so

```

Step 3: The Runner Script (sim.py)

CODE:

```
import sys
from TOSSIM import *
```

```
t = Tossim([])
t.addChannel("LedsC", sys.stdout)
```

```
# Boot mote 1
m = t.getNode(1)
m.bootAtTime(0)
```

```
# Run 20 events
for i in range(20):
    t.runNextEvent()
```

OUTPUT:

```
Activities □ Terminal Wsl03:04 aquib@ubuntu: ~/Tinyos-main/apps/Blink
```

```
[INFO] writing XML schema to app.xml
[INFO] compiling BlinkAppCto object file sim.o
nesc++ -shared -fPIC -o simbuild/micaz/sim.o -g -O0 -gcc-gcc -Wmexic-all -fnesc-include=tos -fnesc-scheduler=TinyScheduler,TinyScheduler.TaskBasic,TaskBasic,TaskBasic,runTask_pos
task -nsec-crfl=tsimbuild/micaz/app.c -fnesc-separator= -DTOSSIM -fnesc-nido-tosnodes=1000 -fnesc-simulate -fnesc-nido-notenubmersim node(\) -fnesc-gcc-gcc -I../././tos/lib/tssn
sim -D_Float128=double -D_Float64x2=double -D_Float64x4=double -I../././tos/platforms/micaz/sim -I../././tos/platforms/micaz/c -I../././tos/platforms/micaz/ca/sim -I../././tos/platforms/mica -I../././tos/platforms/micaz/chips/cc2420/sim -I../././tos/platforms/micaz/chips/cc2420 -I../././tos/chips/cc2420/sim -I../././tos/chips/cc2420 -I../././tos/chips/cc2420/alarm -I../././tos/chips/cc2420/control/sim -I../././tos/chips/cc2420/control -I../././tos/chips/cc2420/csma/sim -I../././tos/chips/cc2420/csma -I../././tos/chips/cc242
0/interfaces/micaz/chips/cc2420/interfaces -I../././tos/chips/cc2420/link/sim -I../././tos/chips/cc2420/link -I../././tos/chips/cc2420/lowpan/sim -I../././tos/chips/cc2420/low
pan -I../././tos/chips/cc2420/lpl -I../././tos/chips/cc2420/packet/sim -I../././tos/chips/cc2420/packet -I../././tos/chips/cc2420/receive/sim -I../././tos/chips/cc2420/receive -I../././to
s/chips/cc2420/spi -I../././tos/chips/cc2420/transmit -I../././tos/chips/cc2420/unique/sim -I../././tos/chips/cc2420/unique -I../././tos/platforms/micaz/chips/at45db -I../././tos/platfo
rm/mica/chips/at45db -I../././tos/chips/at45db -I../././tos/chips/atm128/sim -I../././tos/chips/atm128 -I../././tos/chips/atm128/adc -I../././tos/chips/atm128/pins/sim -I../././tos/chips/atm128/pins -I../././tos/chips/atm128/spi/sim -I../././tos/chips/atm128/spi -I../././tos/chips/atm128/timer/sim -I../././tos/chips/atm128/timer -I../././tos/lib/power -I../././tos/lib/tin
er/firmware/bld/bin/serial -DPLATFORM_MICA2 -Wall -Mehadov -fline-limits=100000 -I../././tos/system -I../././tos/types -I../././tos/interfaces -DIDENT_APPNAME=BlinkAp
pc" -DIDENT_USERNAME=aquib" -DIDENT_HOSTNAME=ubuntu" -DIDENT_USERHASH=0xc8f8622L -DIDENT_TIMESTAMP=0x6ac1fb3l -DIDENT_UIDHASH=0x5c1ddB5L -Wno-nsec-data-race BlinkAppC.nc
-fnesc-dump=components -fnesc-dump=variables -fnesc-dump=constants -fnesc-dump=yedefes -fnesc-dump=interfaces -fnesc-dump=tags -fnesc-dumplib=app.xml
../././tos/lib/tossn/sim_noise.c: In function 'makeNoiseModel':
../././tos/lib/tossn/sim_noise.c:440:78: warning: cast from pointer to integer of different size [-Wpointer-to-int-cast]
dbg("Insert", "Setting history xl to be %ln", (int), (int)noiseData[node_id],key(l));
^
../././tos/lib/tossn/sim_noise.c: In function 'sim_noise_gen':
../././tos/lib/tossn/sim_noise.c:307:7: warning: variable 'noiseIndex' set but not used [-Wunused-but-set-variable]
int noiseIndex = 0;
^
../././tos/lib/tossn/sim_lql.c: At top level:
../././tos/chips/atm128/sim/atm128hardware.h:101:3: warning: '__nesc_disable_interrupt' is static but used in inline function '__nesc_atomic_start' which is not static
__nesc_disable_interrupt();
^
[INFO] compiling Python support and C libraries into pytosstsim.o, tosssim.o, and c-support.o
g++ -c -shared -fPIC -o simbuild/micaz/pytosstsim.o -g -O0 -DIDENT_APPNAME="BlinkAppC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xc8f8622L -DIDENT_T
IMESTAMP=0x6ac1fb3l -DIDENT_UIDHASH=0x5c1ddB5L ../././tos/lib/tossn/tossin_wrap.cxx -I/usr/include/python2.7 -I../././tos/lib/tossin -DHAVE_CONFIG_H
../././tos/lib/tossin/tossin_wrap.cxx: In function 'void SWIG_Python_AddErrorMsg(const char*)':
../././tos/lib/tossin/tossin_wrap.cxx:880:42: warning: format not a string literal and no format arguments [-Wformat-security]
PyErr_Format(PyExc_RuntimeError, msg);
^
g++ -c -shared -fPIC -o simbuild/micaz/tossin.o -g -O0 -DIDENT_APPNAME="BlinkAppC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xc8f8622L -DIDENT_T
IMESTAMP=0x6ac1fb3l -DIDENT_UIDHASH=0x5c1ddB5L ../././tos/lib/tossin/tossin.c -I/usr/include/python2.7 -I../././tos/lib/tossin
g++ -c -shared -fPIC -o simbuild/micaz/c-support.o -g -O0 -DIDENT_APPNAME="BlinkAppC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xc8f8622L -DIDENT_
TIMESTAMP=0x6ac1fb3l -DIDENT_UIDHASH=0x5c1ddB5L ../././tos/lib/tossin/hashtable.c -I/usr/include/python2.7 -I../././tos/lib/tossin
[INFO] linking into shared object _TOSIMModule.so
g++ -shared -fPIC simbuild/micaz/pytosstsim.o simbuild/micaz/sim.o simbuild/micaz/tossin.o simbuild/micaz/c-support.o -lstcd++ -o _TOSIMModule.so
[INFO] copying python script interface TOSSIM.py from lib/tossin to local directory
nesc++
** Successfully built micaz TOSSIM library.
aquib@ubuntu: ~/Tinyos-main/apps/Blink$ python sim.py
0:0:0,488281260 DEBUG (1): LEDS: Led0 off.
0:0:0,976562510 DEBUG (1): LEDS: Led0 on.
aquib@ubuntu: ~/Tinyos-main/apps/Blink$
```